

Wearables and Habits: Project Plan Proposal

Issue #1 — for discussion

3 October 2026

Status: proposal, nothing here has been tested. Statements about the dataset are from memory of the published description and must be confirmed in the data audit.

1 Summary of the proposal

Question: do wearables (fitness trackers, GPS watches) causally increase running? **Data:** Open e-commerce 1.0 (Berke et al., 2024; DOI [10.7910/DVN/YGLYDY](https://doi.org/10.7910/DVN/YGLYDY)), which, as I understand it, contains Amazon purchase histories linked to survey demographics. **Design:** staggered-adoption event study, with the first wearable purchase as treatment.

Central caveat. The data observe *purchases*, not activity. Running can only be measured through proxies (e.g. running shoes, apparel, gear). The research question should therefore be stated as the effect of wearable adoption on *running-related purchasing*, with the link to actual running treated as an assumption, not a result. The second threat is selection: people who decide to start running buy a watch and shoes together, which breaks no-anticipation. The plan is built around these two issues.

2 Questions the plan must answer

Scope

- **Treatment:** first purchase of a wearable (tracker, GPS/sports watch). Smartwatches are a borderline class; proposal is to start with a narrow definition and treat the broad one as robustness.
- **Primary outcome:** running-related purchases (shoes, apparel, accessories) per user-period. **Secondary:** replacement cadence of running shoes as a volume proxy; other fitness categories.
- **Population:** users with an observed wearable purchase (treated) and users with none (never-treated), restricted to users with enough purchase history before and after adoption.
- **Time window:** the full span of the dataset, trimmed by the audit. Out of scope: health outcomes, non-purchase behavior, anything requiring activity data.

Data: sufficient, and what else?

- **Strengths:** individual purchase histories with timestamps and demographics permit event-study timing and heterogeneity by age, sex, income.

- **Weaknesses:** purchases are not use; gifts and household purchases; a single retailer; possibly small treated sample once cohorts are split.
- **Supplements to scope, not to commit to:** aggregate public activity data (e.g. platform trend reports) as external validity for the proxy; a short validation using any source that links purchases to self-reported running (the dataset’s own survey, if it has such items).
- **Alternatives if the audit fails:** different treatment/outcome pair in the same data (e.g. other fitness equipment), or a different dataset with activity logs but weaker identification.

Identification

Element	Proposal
Estimator	Heterogeneity-robust staggered DiD (Callaway–Sant’Anna as primary; Sun–Abraham or de Chaisemartin–D’Haultfoeuille as cross-check). Avoid plain two-way fixed effects.
Comparison group	Never-treated users; not-yet-treated as robustness.
Key assumptions	Parallel trends; no anticipation; stable composition of users.
Tests	Pre-trend event-study plots; placebo outcomes (unrelated categories); placebo treatment (other electronics purchases); sensitivity of trends (HonestDiD-style bounds); dropping users whose running-gear purchases coincide with the wearable within a short window.
Alternatives	Matched DiD on pre-period purchases; heterogeneity cuts. IV is not proposed (no credible instrument known).

Feasibility checks before committing

1. Number of treated users, and how many treatment cohorts (by period).
2. Pre- and post-adoption history length per treated user.
3. Share of users with any running-related purchase; how sparse the outcome is.
4. Whether wearables and running categories can be classified reliably from the product fields.
5. Minimum detectable effect from a simulation using the real treatment timing and outcome variance.

Outputs

A LaTeX report per phase (compiled PDF) under `reports/`; scripts under `code/scripts/`; figures and tables under `code/analysis/outputs/`; a final summary report. A paper in `paper/` is optional and decided at the go/no-go point.

3 Phase outline

#	Phase	Content (headings only)
1	Setup	Container, repo structure, data download with retry/resume, data dictionary.
2	Data audit	Classification of wearables and running products, feasibility checks, power.
GO/NO-GO: after phase 2 (see below).		
3	Design	Fix the treatment and outcome definitions, sample, estimators, and test list before estimating.
4	Analysis	Main event study and heterogeneity.
5	Robustness	Placebos, alternative estimators and treatment definitions, sensitivity.
6	Reporting	Final report, figures, replication notes.

Proposed go/no-go criteria (after the data audit)

Go if all hold: (i) enough treated users with usable pre- and post-periods that the minimum detectable effect is economically meaningful; (ii) wearables and running products can be classified credibly; (iii) pre-trends in a first look are not obviously violated. **Adjust** if only the outcome or treatment definition fails (redefine, re-audit once). **No-go** if (i) fails, then switch data or reframe as descriptive. Numeric thresholds are left to the decision owner, to be set *before* looking at estimates.

4 Open decisions

Decision	Proposed default	Owner	Deadline
Treatment definition	Narrow (trackers, GPS watches)	TBD	TBD
Primary outcome	Running-related purchases	TBD	TBD
Primary estimator	Callaway–Sant’Anna	TBD	TBD
Go/no-go thresholds	Set numerically before estimation	TBD	TBD
Final deliverable	Report series; paper optional	TBD	TBD
Structural extension	Decide only after reduced-form results	TBD	TBD

5 Potential extension (conditional on the reduced-form results)

This section is speculative and is *not* part of the committed plan. It is only opened if the reduced-form evidence clears the gate below.

How broad can the question be?

- **Other habits:** walking/steps, cycling, gym and home-fitness gear, nutrition and sleep products, all visible as purchase categories.

- **Dynamics:** is the effect a durable habit or a novelty spike that decays? Abandonment can be proxied by repurchase or upgrade of the device and by lapses in gear purchasing.
- **Heterogeneity:** who responds (age, sex, income, baseline activity), which matters most for policy.
- **Spillovers and complements:** household members, apparel, accessories, subscriptions.
- **Mechanisms:** self-monitoring/information, goal-setting and gamification, commitment, identity.
- **Policy and welfare:** would subsidies (employer or insurer programs) pay off, and for whom?

Structural approaches that could add something

Model	Idea	What it could answer
Dynamic discrete choice with habit stock	Forward-looking adoption of a durable device; exercise intensity depends on a habit stock the device makes cheaper to build.	Selection versus causal effect of adoption; value of adoption; effect of subsidies.
Learning model	The device gives a signal about the returns to one's own effort; users update, and may quit if the signal is bad.	Why effects fade or persist, and who abandons.
Present bias / commitment	A (partially) naive hyperbolic discounter uses the device as a commitment or self-monitoring tool.	Whether wearables work through self-control, and welfare of targeted subsidies.
Demand with complements (BLP-style)	Joint demand for wearables and running gear with price variation and complementarity.	Cross-price effects, counterfactual prices, spillovers on gear markets.

Link to the reduced form. The event-study coefficients (level, persistence, heterogeneity) become the target moments for indirect inference or simulated-method-of-moments estimation, so the structural model is disciplined by the credible reduced-form evidence rather than replacing it. The `pyblp` package is already in the project container if the demand route is chosen.

Gate for opening the extension

All of the following should hold: (i) a clean, precisely estimated effect on the primary outcome with no obvious pre-trend violation; (ii) enough panel length to see dynamics (persistence or decay); (iii) meaningful heterogeneity worth explaining; (iv) price or product variation sufficient to identify a choice model, if the demand route is chosen. Otherwise the project stays reduced form. Main risk: the outcome is still a purchase proxy, so any structural claim about *running itself* rests on the proxy assumption.

6 Next steps and follow-up issues

1. Review this proposal; resolve the open decisions above.
2. Open one issue per phase: Setup, Data audit, Design, Analysis, Robustness, Reporting.

3. Data audit issue includes the go/no-go memo as its deliverable.